

Fig.2.2(i)

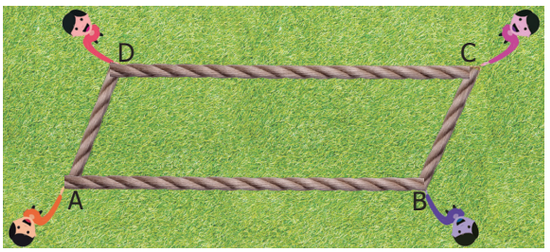


Fig.2.2(ii)

Hence, we can conclude that a parallelogram is a four sided closed shape, in which opposite sides are both parallel and equal.

MATHEMATICS ALIVE - MEASUREMENTS IN REAL LIFE


Wall tiles


Sweet


Window

2.2.1 Area and Perimeter of the Parallelogram

Draw a parallelogram on a graph sheet as shown in Fig.2.3(i) and cut it. Draw a perpendicular line from one vertex to the opposite side. Cut the triangle and shift the triangle to the other side of the parallelogram as shown in Fig.2.3(ii). What shape is seen now? It is a rectangle as in Fig.2.3(iii). Hence, the area of the parallelogram is the same as that of the rectangle.

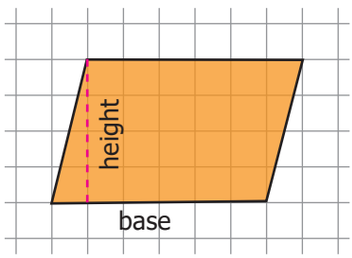
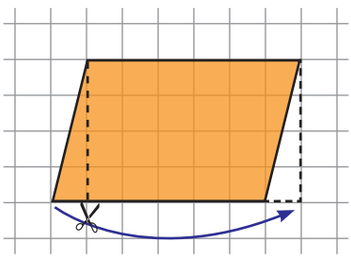


Fig.2.3(i)


Fig.2.3(ii)

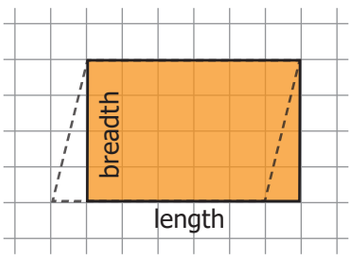


Fig.2.3(iii)

Therefore, area of the rectangle = length x breadth
= base x height $sq.units.$
= b x h $sq.units.$
= Area of the parallelogram.

Further, the perimeter of a parallelogram is the sum of the lengths of the four sides.

Think

1. Explain the area of the parallelogram as sum of the areas of the two triangles.
2. A rectangle is a parallelogram but a parallelogram is not a rectangle. Why?

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